BIRZEIT UNIVERSITY

Faculty of Engineering and Technology
Electrical and Computer Engineering
ENEE3309 Communication Systems, 2023-2024-1st semester Project

---

About the course project:

- General Description:

  This project will cover amplitude modulation and demodulation. A normal AM waveform is produced using a simple switching modulator circuit. The resulting AM waveform is then demodulated using an envelope detector circuit. It requires the students to simulate the circuit and then construct the circuit and monitor the signal in both the time and frequency domain. Plots and discussion are required at each stage to show an understanding of the relevant modulation concepts.

- The projects will be evaluated based on the following deliverables:
  - Project report (of around 3-4 pages)
  - Hardware design
  - Short presentation

- The project submission must be via Ritaj.

- Four of the relevant program outcomes are listed below:

  - **Outcome a:** *"an ability to apply knowledge of mathematics, science, and engineering"*
    The proposed project requires the student to apply communications theory to a practical circuit implementation.
  - **Outcome b:** *"an ability to design and conduct experiments, as well as to analyze and interpret data."*
    The proposed project provides the opportunity for the student to experiment with the circuit parameters and evaluate the circuit response.
  - **Outcome e:** *"an ability to identify, formulate, and solve engineering problems."*
    The proposed project allows the student to solve for several circuit components and signal parameters associated with the assignment.
  - **Outcome k:** *"an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice."*
    The proposed project uses modern simulation software and basic circuit measurement techniques to produce the requested results.

Your report must have the following structure, using these section headings:

Introduction: A general description of the area of your project.

Problem Specification: A clear technical description of the problem. Formulating a general problem (e.g., Normal AM Modulation/Demodulation) into a well-defined general formula.

Data: What are the real-record and/or synthetic signals you are going to use to develop and evaluate your work?

Evaluation Criteria: How are you going to measure how well your project performs? The best criteria are objective, quantitative, and discriminatory. You want to be able to demonstrate and measure improvements in your system.

Approach: A description of how you tried to solve the problem.

Results and Analysis: What happened when you evaluated your system using the data and criteria introduced above? What were the principal shortfalls?

Conclusions: What did you learn from doing the project? What did you demonstrate about how to solve your problem?

References: Complete list of sources you used in completing your project, with explanations of what you got from each.

Phase One (Due on Thu. 4/1/2024):

Part 1: Normal AM modulation.

Normal AM modulation can be generated by using a switching modulator with a bandpass filter.

1.1 Express modulating signal m(t) and carrier signal c(t) in time domain, and frequency domain.

1.2 Assume the diode switching signal $p(t)$ is given by

$$p(t) = f(x) = \begin{cases} 1\,, & -\frac{T_c}{4} \le x \le \frac{T_c}{4} \\ 0, & -\frac{T_c}{2} \le x < -\frac{T_c}{4} \\ 0, & \frac{T_c}{4} \le x < \frac{T_c}{2} \end{cases}$$

Where T_c denotes the fundamental period of carrier signal $c(t)$. Determine the complex exponential Fourier series of the signal $p(t)$.

1.3 Evaluate the output modulated signal $s(t)$.

1.4 Design a suitable bandpass filter to generate normal AM modulation expressed below.

$$\boldsymbol{s(t) = 2\,[1 + 0.8\cos(2\pi 10^3 t)]\cos(2\pi 10^4 t)}$$

1.5 Use MATLAB or Pspice software to plot the modulating signal, carrier signal, switching signal, and modulated signal in the time domain and frequency domain. Explain your results.

Part 2: Normal AM demodulation

To recover the message signal from the modulated signal, use an envelope detector.

2.1 Design the envelope detector to recover the message signal.

2.2 plot the demodulated output signal in both the time domain and frequency domain.

Phase Two (Due on Thu. 18/1/2024): Bread-board Circuit Implementation

Based on your analysis:

- Implement the normal AM modulator and Demodulator breadboard circuit.
- Display the modulated, and demodulated signals on an oscilloscope. Explain your results.